GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. For known prepared positive causal experiments with a finite separating affine family of executable tests, we characterize finite-state realization at an actually executed stopping time. A joint support-function criterion eliminates all report-routing kernels from the first-run occupation equations. Its sufficient direction follows from uniform transience, rather than from a formal balance equation. The resulting exact risk identity separates action regret, preparation mixing, continuing occupation and terminal prediction. Continuing and terminal centroids need not agree when stopping depends on the report. Under actual-row mass and covering certificates and feedback-uniform acquired overlap, we construct a homogeneous observer with one counted preparation label and at most $M-1$ running labels. For a controlled report-dependent stopping hazard $\alpha s_a(y)$, its excess risk and the checkpoint excess are both comparable to $M^{-2/d}$, uniformly as the expected acquisition count grows like α^{-1}. No stationary preparation, resetting operation or hidden posterior register is used. Controlled Gaussian observations and noncommuting quantum instruments verify the same theorem, including stopping on report regions. Actual weighted affine strata, hard-cap approximation, common-task error composition and a fully explicit binary Gaussian program are treated separately.

1. The causal problem

A prediction rule is not determined by the geometry of its final answers. It must acquire reports, select later experiments and retain enough information to execute its next step. At a stopping time there is a further distinction: the law of a state encountered while the experiment is still running is not generally the law of that state when the experiment ends. Selection by the stopping report changes the latter law. A finite-state construction which calibrates only one of these distributions can have the correct one-step formula and still fail to implement its claimed terminal risk.

We start with preparation, positive report instruments and executable future tests. A finite separating affine test closure realizes the predictive quotient. Neither a Fisher matrix nor a covariance geometry is an input. Squared prediction of a fixed protected audit induces the task norm. Its actual acquired laws, rather than the formal mixture domain, determine the compression lower bound.

The first result, Theorem 3.1, is an exact global criterion. It involves finitely many positive occupation vectors, common action probabilities, and explicit inequalities against dual vectors. All report-dependent state-update kernels have been eliminated. Uniform transience identifies a feasible positive solution with the occupation measure of a first run from the given preparation. Thus the criterion is not merely local centroid compatibility, and it does not accept extraneous invariant mass as an acquired law. It also yields attainment of the optimal finite-state risk under the stated finite-dimensional hypotheses.

Theorem 4.1 explains the causal cost of compression. In addition to action regret and the two continuing/terminal Jensen defects, it contains a preparation-mixing defect. Example 4.2 has strictly positive hidden transitions and vanishing local routing defects, but a strictly positive

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preparation defect. Example 4.3 shows why a continuing centroid cannot be used as a terminal decoder when the stop depends on the report. These are failures of specific identifications, not impossibility conclusions about the mother problem.

The main quantitative consequence, Theorem 5.2, allows every full-history adaptive choice from a fixed finite library of sensing and randomized stopping rules. The stop hazard is $\alpha s_a(y)$ after the actual report y. The functions s_a may vanish on report regions. An integrated hazard floor guarantees finite expected acquisition, but the stopping time is not an independent geometric clock. The theorem gives matching checkpoint and homogeneous total-label curves, uniformly in α, by combining two mechanisms. The Bellman oscillation is $O(\alpha)$ under acquired feedback overlap. The continuing occupation has mass $O(\alpha^{-1})$. Their product controls the accumulated nonsmooth curvature loss without an entrance term.

The preparation-injected positive correspondence extends the previously deposited independent-geometric-stop construction [14]. This is a proof construction. The executed machine begins from the specified preparation once; it never resets, starts from an invariant physical ensemble, or stores an exact posterior. A positive resolvent recovers its actual continuing occupation, and a separate terminal pushforward fixes its decoder. These roles are not interchangeable.

Two distinct physical classes verify the same general hypotheses in Section 7. The singular extension retains the actual mass of every marked affine stratum. Section 9 states the stopping, simulator and precision resource semantics, including when numerical errors accumulate with expected acquisition and when a terminal readout error is quadratic. The worked Gaussian program is an explicit finite-budget calculation, not a numerical substitute for the general proofs.

The exact disintegration theorem requires neither a density nor a stability certificate. The matching law additionally requires acquired density, a global legal cover, a buffer for task curvature and acquired feedback overlap. These quantitative conditions are sufficient; the exact criterion does not make them necessary. Occupation balance, Bellman contraction, convex separation and fixed points are classical tools. The contribution claimed here is their preparation-sensitive, terminal-sensitive combination for a fully counted causal observer, together with the uniform stopping/acquisition resource conclusion. The relation to prior work is made explicit in Section 11.

2. Prepared instruments and the stopped task

2.1. **Predictive realization.** Fix a known prepared causal experiment on standard Borel spaces and a finite nonempty action set A. Legal histories contain only previously acquired reports, actions and the observer's independent private randomization. Legal future tests include the constant test, report events and a fixed bounded protected audit, and are closed under the stipulated finite causal continuations. Two histories are equivalent when their legal action interfaces and every executable future-test expectation agree.

Apply the predictive quotient below to live histories with the same acquisition interface. A terminal history belongs to the separate, visibly absorbing interface component, where only the protected audit is legal. Equal physical coordinates do not identify these two interfaces. Assume that finitely many executable coordinates separate the live equivalence classes, that their attained image is Borel with a Borel realization section, and that their unnormalized one-report continuations close affinely. Realizable mixtures give a compact convex base $B \subset \mathbb{R}^d$ with nonempty interior. Put

$$E = \mathbb{R} \oplus \mathbb{R}^d, \quad C_B = \{w(1,u) : w \geq 0,\ u \in B\}, \quad e(w,v) = w.$$

The larger mixture base is not asserted to be the support of any one acquisition law. A compact convex operating domain $D \subset B$ is forward invariant under all instruments considered; it is full dimensional in the chosen test chart. We write C_D for its cone. All positivity and base-norm statements used on an executed ensemble are on this operating cone.

Lemma 2.1 (Measurable predictive quotient)**.** *Under these hypotheses the attained test image realizes the predictive quotient as a Borel causal state space. It is minimal among sufficient Borel history statistics whose attained images have Borel realization sections. Updates descend jointly in the finite action and report coordinates.*

Proof. Equality of coordinates is equivalent to equality of the declared executable tests. The affine continuation identities therefore make the next report law and every next test expectation independent of the representative history. A section of any other sufficient statistic followed by the coordinate map gives its Borel factorization through the test image. The finite action coordinate and common reference-null sets make the descended kernels Borel. Successive integration verifies every finite legal continuation. No assertion that an arbitrary measurable quotient is standard Borel is used. □

2.2. **A marked positive instrument.** For each $a \in A$ let λ_a be a reference probability on a standard Borel report space Y_a. Let $I_a^0(y), I_a^1(y)$ be positive linear maps on C_D, with integrable finite-dimensional operator norms, such that

$$\sum_{b=0}^{1} \int e(I_a^b(y)z)\, d\lambda_a(y) = e(z) \qquad (z \in C_D). \tag{1}$$

The mark $b = 0$ means continue and $b = 1$ means stop. The raw maps, and hence both mark probabilities, are part of the physical experiment. Write

$$I_a^b(y)(1,u) = (\ell_a^b(y,u), U_a^b(y,u)), \qquad F_a^b(y,u) = U_a^b(y,u)/\ell_a^b(y,u).$$

All affine identities hold for every $u \in D$ outside one reference-null set for each action and mark. When $\ell_a^b(y,u) = 0$, positivity on a bounded base forces $I_a^b(y)(1,u) = 0$; no posterior is assigned positive mass there. At positive likelihood $F_a^b \in D$. Define the sub-Markov operators

$$K_a^b f(u) = \int \ell_a^b(y,u) f(F_a^b(y,u))\, d\lambda_a(y). \tag{2}$$

Their measures have their actual masses; no conditioning on successful stopping is silently substituted.

A preparation $u_0 \in D$ is fixed and known. Call indices start at one. The action at call $t+1$ is chosen using the history before that call. After the report and its mark have arrived the observer updates its label. At the first mark one, at time $\tau \geq 1$, it commits a label-only response and the single protected audit is performed. No audit is performed on a continuing run. Private action and update coins are fresh; a retained random seed would be an additional state resource.

We assume *uniform transience*: for some integer $b_0 \geq 1$ and $0 \leq \beta < 1$, every legal full-history policy and every preparation in D satisfy

$$\mathbb{P}_u^\pi(\tau > b_0) \leq \beta. \tag{3}$$

The same condition applies after any continuing history. Consequently $\mathbb{P}(\tau > kb_0) \leq \beta^k$ and $\mathbb{E}\tau \leq b_0/(1-\beta)$. This is a proved stopping-time integrability condition, not a fixed-time statement extended by notation.

2.3. **Task and finite resources.** The actual audit has, in the executable chart, conditional squared-response cost

$$l(u,r) = c(u) - 2\langle u, r\rangle + \|r\|^2, \qquad g(u) = l(u,u) = c(u) - \|u\|^2, \qquad 0 \leq l(u,r) \leq L. \tag{4}$$

Here c is affine and $r \in B$; the bound holds on the operating state and response domains. For a categorical or informationally complete audit, an affine isometric chart gives this formula while retaining the complete audit second moment in c. No sampling-noise component is removed from the physical baseline.

A homogeneous observer has a set S of at most M labels, an initial label s_0, fixed action probabilities p_{sa}, Borel routing probabilities $\gamma_b(j \mid s, a, y)$, and fixed readouts $r_j \in B$. The stop mark is visible before routing and then terminates the physical run. A label may be used both while continuing and at termination; it has one fixed terminal readout. A continuing calibration point and a terminal readout may be two different constants attached to that label. They are program data, not extra dynamic registers. There is no free phase counter or history tape. Let $\mathcal{R}_{\mathrm{hom}}(M; u_0)$ be the infimum of its actual stopped risk.

The full-history comparator may use every legal adaptive acquisition policy and all its history to predict the audit. Its value is $V(u_0) = \inf_\pi \mathbb{E}_\pi g(X_\tau)$, where X_t is the full-history predictive vector on the actual surviving history. A checkpoint permits the same unrestricted acquisition history and, only at the stop, encodes it into M labels with one fixed decoder. Its acquisition workspace is not identified with its terminal label budget.

Expected and hard resources are different. The present interface makes one preparation, τ paid raw acquisitions and one audit. It has finite expected cost $\mathbb{E}\tau + 2$ but no deterministic acquisition cap. A capped interface and the auditor's counter are introduced separately in Corollary 9.2. The stop mark is not an internally timed deterministic deadline.

3. A global stopped-disintegration theorem

For a positive vector $z = w(1, u)$ and a bounded continuous function f on D, write

$$\mathsf{P}_f(z) = wf(u), \qquad \mathsf{P}_f(0) = 0.$$

This perspective is continuous on bounded subsets of C_D. Put $a_s = \mathbf{1}_{\{s=s_0\}}(1, u_0)$ and $T_0 = b_0/(1-\beta)$.

Theorem 3.1 (Global realization from stopped occupation)**.** *In the experiment of Section 2, fix M and use $S = \{1, \dots, M\}$. A collection of continuing occupation vectors $z_s \in C_D$, terminal vectors $d_j \in C_D$, and action rows $p_s \in \Delta(A)$ is realized by a single homogeneous Borel observer if and only if*

$$z_s - a_s \in C_D, \qquad \sum_j e(d_j) = 1, \qquad \sum_s e(z_s) \le T_0, \tag{5}$$

and, for every family of linear functionals $u_j, v_j \in E^*$,

$$\begin{aligned} &\sum_j \langle u_j, z_j - a_j\rangle + \sum_j \langle v_j, d_j\rangle \\ &\qquad \le \sum_{s,a} p_{sa} \int \left[\max_j \langle u_j, I_a^0(y) z_s\rangle + \max_j \langle v_j, I_a^1(y) z_s\rangle\right] d\lambda_a(y). \end{aligned} \tag{6}$$

The continuing vectors have their first-run meaning

$$z_s = \mathbb{E} \sum_{t=0}^{\tau-1} \mathbf{1}_{\{S_t=s\}}(1, X_t), \qquad d_j = \mathbb{E}[\mathbf{1}_{\{S_\tau=j\}}(1, X_\tau)]. \tag{7}$$

In particular,

$$\mathcal{R}_{\mathrm{hom}}(M; u_0) = \min_{(z,d,p)\ \textit{satisfying}\ (5)\text{–}(6)} \sum_j \mathsf{P}_g(d_j). \tag{8}$$

Both minima, in programs and in the displayed positive vectors, are attained. A tuple violating (6) *has a strict finite-dimensional affine separation witness. No density, geometric cover, stationary initial law or mixing hypothesis is required.*

The criterion is global: it enforces both preparation-injected continuing flow and the terminal law using one physical program. It is not a list of independent one-step quantization minima. It is a finite-dimensional, generally nonconvex optimization with a family of dual inequalities, not a polynomial-time synthesis result. The action probabilities and occupation vectors occur multiplicatively. We have not interchanged an outer minimization with the support-function maximization.

Proof. For fixed routing kernels define positive linear ensemble maps on C_D^M by

$$(Pz)_j = \sum_{s,a} p_{sa} \int \gamma_0(j \mid s,a,y) I_a^0(y) z_s \, d\lambda_a(y),$$

$$(Qz)_j = \sum_{s,a} p_{sa} \int \gamma_1(j \mid s,a,y) I_a^1(y) z_s \, d\lambda_a(y).$$

Equation (1) gives $e_{\mathrm{tot}}(Pz) + e_{\mathrm{tot}}(Qz) = e_{\mathrm{tot}}(z)$ on the positive cone. Uniform transience gives $e_{\mathrm{tot}}(P^{b_0} z) \le \beta e_{\mathrm{tot}}(z)$. Positive maps with this bound contract the associated base norm by at most β every b_0 powers. Thus

$$(I-P)^{-1} = \sum_{t \ge 0} P^t \tag{9}$$

converges in finite-dimensional operator norm, with norm at most T_0.

Starting from $a = (a_s)$, repeated conditional integration of the raw instrument gives the actual unnormalized ensemble $P^t a$ at continuing cut t. Summation proves that its occupation is $z = (I-P)^{-1}a$ and that its terminal ensemble is $d = Qz$. These are exactly (7). The mass of d is one, since the probability of indefinite continuation is zero. The other conditions in (5) follow. Routing into finitely many labels is a convex combination of the displayed dual values at every (s,a,y,b); its maximum bounds that combination. This proves necessity of (6).

For sufficiency fix a tuple (z,d,p). Let each $\gamma_b(\cdot \mid s,a,\cdot)$ range over the probability-valued unit ball in $L^\infty(\lambda_a)^M$. Positivity and the sum-to-one condition are weak-star closed. The finite product of these sets is compact and convex. The map

$$\gamma \longmapsto (Pz, Qz)$$

is affine and weak-star continuous into a finite-dimensional space, because every coordinate of $I_a^b(y) z_s$ is integrable. Its image $\mathcal{K}(z,p)$ is consequently compact and convex. Its support function at (u,v) is the right side of (6). Indeed one may choose the least maximizing label at each report and mark, a measurable selection from a finite set. Null-set versions can be made Borel and completed arbitrarily; no actual report assigns mass to the modified reference-null sets.

Finite-dimensional separation now implies $(z-a,d) \in \mathcal{K}(z,p)$ if and only if (6) holds for all (u,v). Choose a witnessing pair of Borel routing kernels. Then $z = a + Pz$ and $d = Qz$. By (9), the first equation has exactly one solution, namely the actual first-run occupation. Hence the candidate d is also the actual terminal ensemble. This step is essential: a balance equation without a transience or other identification condition can contain extraneous invariant mass, as the absorbing-control literature emphasizes [3, 4].

For any fixed terminal ensemble, completing the square gives

$$\sum_j \left[\overline{c}(d_j) - 2\langle (d_j)_{\mathrm{pred}}, r_j \rangle + e(d_j)\|r_j\|^2 \right] = \sum_j \mathsf{P}_g(d_j) + \sum_{j : e(d_j) > 0} e(d_j)\|r_j - \bar{r}_j\|^2,$$

where $\bar{r}_j = (d_j)_{\mathrm{pred}}/e(d_j)$ and $\overline{c}(w, wu) = wc(u)$. Each positive-mass centroid lies in $D \subset B$, so it is a legal decoder. A zero-mass decoder is arbitrary. This proves (8).

Finally (5) confines z,d,p to a compact finite-dimensional set. Each support-function inequality is closed: changing z,p changes its right side continuously by the integrable operator-norm bound. The objective is continuous, including at zero mass, since g is bounded and continuous. The feasible

set is nonempty, because any constant-label legal observer has finite expected lifetime by (3). A minimum exists. The sufficient direction realizes its minimizer by a Borel program. Separation of a tuple outside the compact convex routing image gives the asserted strict witness. □

Remark 3.2 (The report interface is part of the theorem). If routing must be chosen before the stop mark is revealed, the same kernel must be used for both marks. Then (6) is replaced by

$$\sum_{s,a} p_{sa} \int \max_j \{\langle u_j, I_a^0(y) z_s\rangle + \langle v_j, I_a^1(y) z_s\rangle\}\, d\lambda_a(y)$$

on the right. A sum of two maxima would wrongly grant two different programs. The support-function proof gives the altered criterion directly. The constructive observer in Theorem 5.2 uses one common routing partition for both marks and therefore satisfies even this stronger interface.

Corollary 3.3 (One program for a finite model family)**.** *For a finite family of raw instruments on the same finite test cone and interface, assume a common transience bound. Require the same p and routing kernels for all models θ. The exact feasibility criterion has vectors (z^θ, d^θ) for every model. In* (6) *sum the left side over θ, and replace each right-side maximum by*

$$\max_j \sum_\theta \langle u_j^\theta, I_{a,\theta}^0(y) z_s^\theta\rangle, \qquad \max_j \sum_\theta \langle v_j^\theta, I_{a,\theta}^1(y) z_s^\theta\rangle,$$

with the appropriate common dominating reference. Terminal readouts must also be common when optimizing a minimax task. Separate modelwise maxima do not characterize a parameter-independent program.

Proof. Take the joint finite-dimensional image of the same weak-star compact routing domain. Its support function places the model sum inside each label maximum. Separation and the modelwise positive resolvents prove equivalence. Compactness also proves attainment of the common-readout minimax problem, since its objective is the maximum of finitely many continuous actual risks. This is common-program robustness, not learning an unknown kernel; the quantitative theorem below concerns a known experiment. □

4. The preparation and stopping defects

The finite-state optimization admits a useful exact decomposition. Let V solve the stopped Bellman equation

$$V(u) = \min_{a\in A}\{K_a^1 g(u) + K_a^0 V(u)\}. \tag{10}$$

Under Section 2, this continuous concave solution exists, is bounded between zero and L, and equals the full-history value. To see this, finite-step iteration gives the stopped problem with a zero residual payoff. The b_0th iterate is a sup-norm contraction with coefficient β, by uniform transience. Positivity and the perspective of a concave function preserve concavity, as does the minimum over the finite action set. Uniform convergence proves the claims. Bellman verification along any legal history leaves a remainder at most $L\mathbb{P}(\tau > n)$, which vanishes. A Borel minimizing selector attains the value. Thus stopping is handled by a convergent stopped argument, not by a formal infinite-horizon extrapolation.

For any legal homogeneous observer use its actual occupation z, terminal ensemble d and continuing arrivals $y = Pz = z - a$. At positive masses write

$$q_s = \frac{(z_s)_{\mathrm{pred}}}{e(z_s)}, \qquad q_j^+ = \frac{(y_j)_{\mathrm{pred}}}{e(y_j)}, \qquad r_j^* = \frac{(d_j)_{\mathrm{pred}}}{e(d_j)}.$$

These are three different conditional means. None is asserted to equal the mean at every deterministic time. Define finite measures indexed by the incoming state, action, mark and outgoing state by

$$d\eta_{saj}^b(y) = e(z_s) p_{sa} \gamma_b(j \mid s, a, y) \ell_a^b(y, q_s)\, d\lambda_a(y). \tag{11}$$

In integrals against this measure put $W = F_a^b(y, q_s)$. Zero rows contribute zero.

Theorem 4.1 (Exact stopped causal defect identity)**.** *For every such observer, with its actual fixed readouts r_j,*

$$R - V(u_0) = \mathcal{A} + \mathcal{J}_{\mathrm{prep}} + \mathcal{J}_{\mathrm{cont}} + \mathcal{D}_{\mathrm{term}}, \tag{12}$$

where each term is nonnegative and

$$\mathcal{A} = \sum_{s,a} e(z_s) p_{sa} \big[K_a^1 g(q_s) + K_a^0 V(q_s) - V(q_s)\big], \tag{13}$$

$$\mathcal{J}_{\mathrm{prep}} = \sum_s \mathsf{P}_V(z_s) - V(u_0) - \sum_j \mathsf{P}_V(y_j), \tag{14}$$

$$\mathcal{J}_{\mathrm{cont}} = \sum_j \mathsf{P}_V(y_j) - \sum_{s,a,j} \int V(W)\, d\eta^0_{saj}, \tag{15}$$

$$\mathcal{D}_{\mathrm{term}} = \sum_{s,a,j} \int \|W - r_j^*\|^2 \, d\eta^1_{saj} + \sum_j e(d_j) \|r_j - r_j^*\|^2. \tag{16}$$

The preparation defect vanishes when the initial label is never revisited and all other labels have no preparation injection. Local zero routing defects alone do not imply global losslessness.

Proof. The measures in (11) are the actual cross-time report submeasures. The raw affine identities commute with their finite occupation sum: for example,

$$\sum_t \mathbb{E}\big[\mathbf{1}_{\{t<\tau, S_t = s\}} I_a^b(y)(1, X_t)\big] = I_a^b(y) z_s.$$

This is an identity of unnormalized integrable vectors, not an interchange of normalization and conditional expectation. Fresh action and routing coins depend on the physical state only through the declared label and report. Hence y_j and d_j are the barycentric integrals of $(1, W)$ against the corresponding measures.

The Bellman inequality makes (13) nonnegative. The balance $z_s = a_s + y_s$ and concavity of the perspective make (14) nonnegative. Jensen's inequality applied separately to each continuing arrival makes (15) nonnegative. Formula (16) is nonnegative by inspection. Because c is affine, completing the square within every terminal label gives

$$R = \sum_{s,a} e(z_s) p_{sa} K_a^1 g(q_s) + \mathcal{D}_{\mathrm{term}}.$$

Expanding the three other displayed terms cancels all continuing integrals and all occupation potentials, leaving the first sum minus $V(u_0)$. This proves (12). Every integral is finite by boundedness and uniform transience. If the preparation label has $y_{s_0} = 0$ and all other $a_s = 0$, the perspective contributions in (14) cancel individually. □

Example 4.2 (Vanishing local defects do not remove preparation mixing)**.** Consider a two-state hidden chain, no informative report, and transition matrix

$$P = \begin{pmatrix} 1/4 & 3/4 \\ 3/4 & 1/4 \end{pmatrix}.$$

All its entries and its sole report likelihood are strictly positive. Prepare state zero with certainty, and after each transition stop with probability $1/2$, independently of that call's hidden state. The terminal audit is its Bernoulli state, scored by scalar squared loss. In the belief coordinate u, $F(u) = 3/4 - u/2$ and $g(u) = u(1-u)$. The full-history value is

$$V(u) = \tfrac{1}{4} - \tfrac{1}{7}(u - \tfrac{1}{2})^2,$$

since $\mathbb{E}(1/4)^\tau = 1/7$. A one-label observer has continuing occupation mass two and mean $q = 3/10$; its continuing arrival and terminal means are both $3/5$. With decoder $3/5$, its risk is $6/25$, whereas

$V(0) = 3/14$. There is only one action, every local acquired row is a point mass, and both routing defects vanish. Nevertheless

$$R - V(0) = \mathcal{J}_{\mathrm{prep}} = 2V(3/10) - V(0) - V(3/5) = \frac{9}{350} > 0.$$

The preparation defect is therefore necessary even for strictly positive transition models. A formal stationary calibration would have erased the source of this difference.

Example 4.3 (Continuing and terminal calibration separate)**.** At every call prepare a fresh fair hidden bit and report it through a binary symmetric channel of accuracy $3/4$. All report probabilities are positive. Stop with probability $3/4$ after report one and with probability $1/4$ after report zero. The pre-mark posterior is $3/4$ or $1/4$, each with mass $1/2$. Conditional on continuation its mean is $3/8$; conditional on stopping it is $5/8$. Using the continuing mean as the terminal response adds exactly $1/16$ to the squared risk. This remains the calculation at the eventual stop, since each fresh call has the same kernel. Independent geometric stopping would conceal this distinction.

Remark 4.4 (Why terminal mass needs its own certificate)*.* If unrestricted acceptance rules are admitted without a uniform acquisition or hazard condition, even an ordinary full-dimensional row need not give a uniform terminal small-ball law. With independent uniform reports $X \in [-1, 1]$, stop on $|X| \le h$ and use the one-label response zero. Then $\mathbb{E}\tau = 1/h$ and terminal squared prediction risk is $h^2/3$. The ordinary row did not change, but the actually stopped law did. This example locates the missing hypothesis; it does not prohibit a different acquisition–memory theorem for that task. The quantitative class below pays for a uniform integrated hazard floor and keeps the stopping-induced density factor explicitly.

5. Acquired geometry under controlled report-dependent stopping

We now give experiment-level conditions under which the exact criterion has matching resource consequences. A normalized positive acquisition instrument $L_a(y)$ is given before the stopping mark. Put

$$L_a(y)(1, u) = (\ell_a(y, u), U_a(y, u)), \quad F_a(y, u) = U_a(y, u)/\ell_a(y, u), \quad \mu_{a,u} = F_a(\cdot, u)_{\#}(\ell_a(\cdot, u)\lambda_a).$$

Write $K_a f = \int f \, d\mu_{a,u}$. These are actual one-call acquired laws. Finite uncensored core experiments are available for verifying the next overlap certificate; they are not extra calls supplied to the observer in the stopped task.

Definition 5.1 (Regular acquired class and a hazard library)**.** The normalized raw maps are positive and integrable on the physical test cone. There are compact convex full-dimensional sets K, D with $K + B_r \subset D \subset B$ for some $r > 0$, and every normalized raw output from D lies in K. Off common null sets the likelihood is positive on D. On that domain require

$$\mu_{a,u} \le \Lambda \operatorname{vol}_d|_K, \qquad \mathrm{TV}(\ell_a(\cdot, u)\lambda_a, \ell_a(\cdot, v)\lambda_a) \le L_{\mathrm{rep}}\|u - v\|. \tag{17}$$

The rows are weakly continuous. A common grid on the full declared K, not just on a selected stationary support, has at most m convex cells with diameter $C_K m^{-1/d}$ and null row boundaries. For compact convex K the displayed density bound supplies such grids.

For some integer $b \ge 1$, $\epsilon > 0$ and probability ζ on K, every legal b-call full-history feedback policy for the normalized core satisfies

$$\mathrm{Law}_u^{\pi}(X_b) \ge \epsilon\zeta \qquad (u \in D). \tag{18}$$

This is a lower bound on a real subprobability component of every such execution, not a latent-state mixing assertion.

Each action includes a prescribed Borel hazard function $s_a : Y_a \to [0, 1]$. Put

$$D_a f(u) = \int s_a(y)\ell_a(y, u) f(F_a(y, u)) \, d\lambda_a(y), \qquad 0 < \eta_- \le D_a 1(u) \le \eta_+ \le 1. \tag{19}$$

The bounds hold for every $a, u \in A \times D$. No pointwise positive lower bound on $s_a(y)$ is required. Fix $0 < \alpha \leq 1$ and define the actual marked instruments

$$I_a^1(y) = \alpha s_a(y) L_a(y), \qquad I_a^0(y) = (1 - \alpha s_a(y)) L_a(y). \tag{20}$$

The stop decision is generated after the report using this physical kernel. The action, including its hazard rule, is selected before the report.

These assumptions precede the observer and its budget. They do not presuppose a stationary distribution, a calibrated finite filter or mixing of that filter. The law of the terminal state is in general different from the raw row and from the normalized continuing occupation. A report region with $s_a = 0$ is an allowed support change in the marked experiment. The hazard lower bound is on its actual integrated mass.

Theorem 5.2 (Uniform stopped acquired-geometry transfer)**.** *Assume Definition 5.1, the executable quotient and the protected task of Section 2. For every known $u_0 \in D$, $0 < \alpha \leq 1$ and $M \geq 2$, there is a homogeneous Borel observer with at most M total labels, using one preparation label and at most $M - 1$ common-grid labels, such that*

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{cp}}(\alpha, M; u_0) - V_\alpha(u_0) \leq CM^{-2/d}, \tag{21}$$

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{hom}}(\alpha, M; u_0) - V_\alpha(u_0) \leq CM^{-2/d}. \tag{22}$$

The constants are independent of α, M, u_0. They depend on the fixed raw, task, density, covering, buffer, hazard and overlap certificates. The benchmark optimizes every legal full-history adaptive choice from the same sensing/hazard library; it is not a fixed-controller comparison.

The value V_α is the bounded concave solution of

$$V_\alpha = \min_a \{\alpha D_a g + (K_a - \alpha D_a) V_\alpha\}. \tag{23}$$

Every legal policy obeys

$$\frac{1}{\alpha \eta_+} \leq \mathbb{E}\tau \leq \frac{1}{\alpha \eta_-}, \qquad \mathbb{P}(\tau > t) \leq (1 - \alpha\eta_-)^t. \tag{24}$$

Thus these are jointly uniform label and finite-expected-acquisition laws. The constructed table may depend on the known preparation, the experiment, M and α. It does not depend on an inaccessible true parameter or on the eventual stopping time.

The continuing centroids and terminal decoders are separately calibrated to the actual first run. The program uses no resetting operation, stationary physical initialization, exact posterior register or retained time index. The routing can be selected before observing the stop mark. In the exact characterization of Theorem 3.1, its value satisfies

$$\Phi_{\mathfrak{E}}(\alpha, M; u_0) = \mathcal{R}_{\mathrm{hom}}(\alpha, M; u_0), \qquad \Phi_{\mathfrak{E}} - V_\alpha \asymp M^{-2/d}.$$

For a prescribed expected raw budget $N \geq \eta_-^{-1}$, choosing $\alpha = (\eta_- N)^{-1}$ guarantees $\mathbb{E}\tau \leq N$ and $\mathbb{E}\tau \geq (\eta_-/\eta_+)N$ for every legal policy in this specified interface. The comparator remains that same hazard-scaled experiment, not every stopping rule satisfying only an expected-time constraint.

The risk curve concerns a controlled dynamic prediction task. It is not a static-parameter sample complexity law. The expected acquisition count is of order α^{-1}; the same expression is not a hard cap. The last display connects the global finite-state feasibility criterion to actual acquired geometry, rather than identifying a supplied coordinate dimension with information that has not been acquired.

Lemma 5.3 (Stopping sensitivity of the Bellman value)**.** *Under Definition 5.1,*

$$0 \leq V_\alpha \leq L, \qquad \mathrm{osc}_D V_\alpha \leq \frac{2b\eta_+ L}{\epsilon}\alpha, \qquad \mathrm{Lip}_K V_\alpha \leq \frac{2b\eta_+ L}{\epsilon r}\alpha. \tag{25}$$

Proof. The probability of continuation after each live history is between $1-\alpha\eta_+$ and $1-\alpha\eta_-$. Multiplication of these conditional bounds proves the two tails and the expectation inequalities in (24). The killed Bellman operator is a contraction in sup norm with coefficient $1-\alpha\eta_-$. Concave perspectives and the minimum over finite actions preserve concavity. Iteration and the stopped verification argument preceding Theorem 4.1 prove existence, concavity, the value interpretation and $0 \le V_\alpha \le L$.

Let $\mathcal{T}_0 f = \min_a K_a f$. It is sup-norm nonexpansive and fixes constants. For any f, q, comparison using the same b-step policy on each side and (18) gives

$$\operatorname{osc}_D(\mathcal{T}_0^b f - \mathcal{T}_0^b q) \le (1-\epsilon)\operatorname{osc}_D(f-q).$$

Indeed all possible terminal expectations of $f-q$ lie between $\epsilon\zeta(f-q)+(1-\epsilon)\inf(f-q)$ and the analogous supremum. This comparison allows different minimizing feedback actions, not just a fixed controller.

Equation (23) implies

$$\|V_\alpha - \mathcal{T}_0 V_\alpha\|_\infty \le \alpha \sup_a \|D_a(g - V_\alpha)\|_\infty \le \alpha\eta_+ L.$$

Telescoping b nonexpansive iterates and then taking oscillations yields

$$\operatorname{osc} V_\alpha \le 2b\alpha\eta_+ L + (1-\epsilon)\operatorname{osc} V_\alpha.$$

This proves the oscillation bound. For a concave function on D, extending a line through two points of K by length r at either end bounds each intermediate slope by $\operatorname{osc}_D V_\alpha / r$. This gives the Lipschitz bound without differentiability or a Hessian of the Bellman minimum. Neither a pathwise filter contraction nor observer mixing was used. □

Lemma 5.4 (Actual-mass curvature bound)**.** *If f is concave and L_f-Lipschitz on compact convex $K \subset \mathbb{R}^d$ and μ is a finite measure with density at most A, a common grid $\mathcal{P}_m$ with at most m cells satisfies*

$$\sum_{C\in\mathcal{P}_m} \mu(C) f(\bar{x}_C) - \int f\, d\mu \le C_{K,d} A L_f m^{-2/d}. \tag{26}$$

Zero-mass terms are zero. The grid is independent of f, μ; its conditional centroids are not.

Proof. We recall the boundary-valid curvature argument to keep the new upper independent of a companion. The convex function $-f$ extends to a convex L_f-Lipschitz function on $\mathbb{R}^d$ by taking the supremum of its interior supporting planes. Its distributional Laplacian κ is a positive locally finite measure. On a cube of side s, subtract a supporting affine function at its center after mollification. The remaining nonnegative convex function w vanishes there. Convexity on radial segments gives $w(z+R\theta) \le R\partial_R w(z+R\theta)$. A submean bound on the cube, followed by integration over the sphere of radius $R = 2\sqrt{d}\, s$ and the divergence theorem, gives

$$\sup_C w \le C_d s^{2-d} \kappa(\overline{B}_{3\sqrt{d}\, s}(z)).$$

For $d=1$ the same statement follows from the two endpoint slopes. Affine terms cancel at the conditional centroid; its value of w is nonnegative. Hence the cell Jensen loss is at most the last bound times $\mu(C) \le As^d$. The enlarged balls have bounded overlap. A smooth cutoff on a fixed neighborhood of K bounds their total curvature mass by $\int \chi\, d\kappa = -\int \nabla\chi\cdot\nabla(-f) \le L_f \int |\nabla\chi|$. Summing and passing through mollification proves a bound $C_{K,d} A L_f s^2$. Subdivide a bounding cube $\lfloor m^{1/d} \rfloor$ times in each coordinate, intersect with K and discard empty cells. Its cell count is at most m and its mesh has the required order. The fixed small cases are absorbed in the constant. No interior mass cutoff or smooth Bellman surface is assumed. □

6. First-run calibration and the matching converse

Lemma 6.1 (Preparation-injected continuing calibration)**.** *For every common grid $\{C_j\}_{j=1}^m$ of Definition 5.1, there exists a homogeneous observer on labels $0, 1, \dots, m$ with the following properties. Label zero is the fixed preparation label and is never revisited. Each running representative q_j lies in $\overline{C}_j$, each action row minimizes the stopped Bellman objective at that representative, and both mark branches route y to the cell containing $F_a(y, q_s)$. The representatives are the normalized means of the actual continuing occupation. Its terminal decoder is, separately, the centroid of the actual terminal measure in the same cell.*

Proof. Use the compact convex set of normalized positive ensembles

$$\mathcal{E}_m = \left\{ (z_0, \dots, z_m) : z_0 = t(1, u_0),\ t \geq 0,\ z_j \in C_{\overline{C}_j},\ \sum_{j=0}^{m} e(z_j) = 1 \right\}.$$

At a positive-mass row use its normalized mean q_s; put $q_0 = u_0$ regardless of its mass. Choose any action probability supported on minimizers of $\alpha D_a g(q_s) + (K_a - \alpha D_a)V_\alpha(q_s)$. At zero running mass choose any representative in its cell; its contribution is zero.

Send each continuing raw contribution into its common cell. Put

$$Y_j(z, p) = \sum_{s,a} p_{sa} \int (1 - \alpha s_a(y)) \mathbf{1}_{\{F_a(y,q_s) \in C_j\}} L_a(y) z_s \, d\lambda_a(y),$$

$$t'(z, p) = \sum_{s,a} p_{sa} \int \alpha s_a(y) e(L_a(y) z_s) \, d\lambda_a(y).$$

The correspondence has output $z_0' = t'(1, u_0)$ and $z_j' = Y_j$. It maps $\mathcal{E}_m$ into itself, since actual continuing normalized outputs in a cell have their barycenters in that cell and total lost mass is exactly t'. In particular $\alpha\eta_- \leq t' \leq \alpha\eta_+$.

Its values are nonempty compact and convex. For fixed z the action-minimizer sets are finite simplices, and the displayed outputs are linear in their product probabilities. Its graph is closed. At a positive limiting mass, normalized means converge; continuity of the Bellman objectives preserves the minimizing action support. The normalized raw maps are continuous in u off common null sets. A cell boundary has zero actual row mass, and positive likelihood makes its preimage reference-null. Dominated convergence therefore handles the cell indicators, even though $s_a(y)$ need only be Borel. At a zero limiting mass the integrable operator bound makes the whole row contribution vanish uniformly in its possible cell representatives. No division by zero or tie-breaking continuity is required.

Kakutani's finite-dimensional theorem [5] supplies a fixed point z^*. Fix its witnessing action rows and all zero-row representatives. Let P be the continuing ensemble map of this fixed physical program, and $a_0 = (1, u_0)$, $a_j = 0$ for $j > 0$. The fixed-point equations say

$$z^* = Pz^* + t^* a, \qquad t^* = e(z_0^*) > 0.$$

The hazard floor implies $\|P\| \leq 1 - \alpha\eta_-$ in the operating base norm. Therefore

$$z^*/t^* = (I - P)^{-1} a. \tag{27}$$

This is the actual continuing occupation from the single given preparation, not a stationary initialization. Its total mass is $1/t^* = \mathbb{E}\tau$. The injection used in the fixed-point proof is never executed as a reset.

For each terminal label take the actual terminal vector

$$d_j = \sum_{s,a} p_{sa} \int \alpha s_a(y) \mathbf{1}_{\{F_a(y,q_s) \in C_j\}} L_a(y) (z_s^*/t^*) \, d\lambda_a(y).$$

It has total mass one by (27). Its positive-mass mean r_j^* lies in $\overline{C}_j$ and is the fixed terminal readout. In general it is not q_j. The program attaches both numbers to the same label as separately charged constants. The common routing rule ignores the stop mark, so it is valid even when that mark is revealed only after the label update. All dynamic memory consists of the $m+1$ labels. □

Proof of Theorem 5.2. Take $m = M-1$ in Lemma 6.1. The preparation label is exclusive, so $\mathcal{J}_{\mathrm{prep}} = 0$. Bellman-minimizing action rows give $\mathcal{A} = 0$. The terminal readouts are their actual centroids. Thus Theorem 4.1 reduces the excess to the continuing Jensen loss plus terminal squared distortion.

The actual aggregate continuing row measure, before assigning its cells, has density at most $\Lambda\mathbb{E}\tau \le \Lambda/(\alpha\eta_-)$, since it is the sum of continuing submeasures of actual rows. Its cell centroids are the running q_j. Lemmas 5.4 and 5.3 give

$$\mathcal{J}_{\mathrm{cont}} \le C_{K,d}\frac{\Lambda}{\alpha\eta_-}\frac{2b\eta_+ L\alpha}{\epsilon r}m^{-2/d} \le Cm^{-2/d}.$$

The factor α in the Bellman oscillation cancels the actual occupation factor α^{-1}. This is why there is no initial-distribution entrance term. The terminal distortion is at most the squared cell diameter, because its actual total mass is one. This proves the upper in (22). Since $m = M-1 \ge M/2$ for $M \ge 2$, its order is as displayed.

For the converse, consider any full-history acquisition policy π in the same marked experiment. Its actual terminal predictive law is

$$\nu_\alpha^\pi(B') = \mathbb{E}_\pi \sum_{t=0}^{\tau-1} \alpha \int s_{A_t}(y)\ell_{A_t}(y, X_t)\mathbf{1}_{\{F_{A_t}(y,X_t)\in B'\}} d\lambda_{A_t}(y). \tag{28}$$

The occupation form is essential; conditioning each summand to have mass one would be wrong. Equation (17) and (24) give

$$\nu_\alpha^\pi \le (\Lambda/\eta_-)\,\mathrm{vol}_d\,|_K. \tag{29}$$

For a probability with density at most A, the union of radius-r_0 balls about M codepoints has mass at most $AMv_d r_0^d$. Choose $r_0 = (2AMv_d)^{-1/d}$. At least half the actual mass lies outside, so

$$q_M(\nu) \ge \tfrac{1}{2}(2Av_d)^{-2/d}M^{-2/d}. \tag{30}$$

Off-support codepoints and randomized encoders do not improve the pointwise minimum distance.

Conditioning the protected squared task on its full stopped history gives the exact checkpoint identity

$$\mathcal{R}_{\mathrm{cp}}(\alpha, M; u_0) = \inf_\pi \left\{\mathbb{E}_\pi g(X_\tau) + q_M(\nu_\alpha^\pi)\right\}. \tag{31}$$

The decoder has no additional time argument. Uniformly over π, (29) supplies the lower $cM^{-2/d}$, while a global grid of K supplies an upper $CM^{-2/d}$. Choose a full-history Bellman minimizing policy for the upper. A finite observer's existing label is a permitted checkpoint encoding of its stopped history and private coins. Therefore the same lower applies to every legal homogeneous observer. Together with the constructed upper, this proves both matching curves.

The converse can even allow time-dependent finite-state tables: at each deterministic terminal time the last stopped row, conditioned on that time, has density bounded by Λ/η_-, because its unnormalized mass is at least $\alpha\eta_-$ times the preceding survival mass. Apply the finite-codepoint lower at that time and average. This is only a larger lower-bound class; no free clock has been added to the constructed homogeneous observer. □

7. Two raw instrument realizations

The hypotheses below are verified before a finite observer is chosen. In particular, a calibration table is not used to establish its own acquired-density hypothesis.

Theorem 7.1 (Controlled Gaussian observations with report-selected stopping)**.** *Let a hidden state take $q \geq 2$ values. After action a it emits a report with density f_i^a in $\mathbb{R}^{q-1}$, conditional on state i, and then undergoes a fixed invertible column-stochastic transition P whose entries are at least $p_* > 0$. Each f_i^a is the density of $N(m_i^a, \Sigma_a)$, where Σ_a is positive definite and the $q-1$ vectors $m_i^a - m_q^a$ are linearly independent. The finite action library may contain different such sensors. The raw instrument, displayed as a density with respect to Lebesgue measure, is*

$$L_a(y)x = P\,\mathrm{diag}(f_1^a(y), \dots, f_q^a(y))x. \tag{32}$$

Let s_a be the indicator of a Borel report region E_a with positive Lebesgue measure, or any fixed Borel function in $[0,1]$ whose Gaussian integral is positive under each hidden state. The physical stop probability is $\alpha s_a(y)$. With a categorical protected audit of the post-transition state, Theorem 5.2 applies in dimension $d = q-1$. Its constants are uniform in α, M and the known preparation in the declared domain. No Gaussian tail is removed.

Proof. To use the reference-probability convention of Section 2, take density $h_a = q^{-1}\sum_i f_i^a$ for λ_a and divide the displayed map by h_a. This leaves every actual law unchanged and gives integrable operator norms. The physical base is the probability simplex. Categorical audit coordinates, with one redundant coordinate removed isometrically, separate all mixtures. Their unnormalized continuations are precisely (32); this proves finite executable affine closure, rather than inferring it from a filter formula. Put $K = P\Delta_{q-1}$ and choose a slightly enlarged convex domain D with $K + B_r \subset D \subset \operatorname{int}\Delta_{q-1}$. Every raw output from the full physical simplex lies in K. The reference likelihood is everywhere positive, and the report law is Lipschitz in the incoming mixture since $\mathrm{TV}(\sum_i u_i f_i^a, \sum_i v_i f_i^a) \leq \frac{1}{2}\|u - v\|_1$.

Write $W_i = u_i f_i^a(Y)/\sum_j u_j f_j^a(Y)$ for the posterior before transition. For $i < q$,

$$Z_i = \log(W_i/W_q) = \log(u_i/u_q) + \log(f_i^a(Y)/f_q^a(Y)).$$

The second term is affine in Y, with nonsingular linear matrix by the stated likelihood-rank hypothesis. Under the actual mixture report law, Z is a finite mixture of nonsingular Gaussians. Their means and inverse covariances are uniformly bounded for $u \in D$ and finite a, and their covariance determinants have a uniform positive lower bound. Thus their densities are bounded by $C\exp(-c\|Z\|^2 + C\|Z\|)$. The inverse logistic Jacobian is $\prod_{i=1}^q W_i^{-1}$ and grows at most exponentially in $\|Z\|$. Its product with this Gaussian tail has a finite supremum on all $\mathbb{R}^{q-1}$. The density of W is therefore bounded on the *entire* open simplex, including limits to its boundary. The invertible affine map P gives a uniform acquired density on K. Its Jacobian determinant and the likelihood-rank determinant are among the fixed constants.

For overlap choose a small closed simplex ball $G \subset \operatorname{int}\Delta_{q-1}$. The same log-odds density and Jacobian are continuous and strictly positive on the compact set $D \times G$ for each finite a. Their minimum is positive. After applying P, every actual one-call row dominates $\epsilon\zeta$, where ζ is normalized volume on PG. This proves (18) with $b = 1$, even when the action is selected from a previous history. Global convex grids, null boundaries and weak continuity follow from the density and dominated integration.

Finally

$$D_a 1(u) = \sum_i u_i \int s_a(y) f_i^a(y)\,dy \geq \min_{a,i} \int s_a f_i^a > 0.$$

This is an integrated bound. The stop can have zero probability on a half-space or another report region. The stopped-law theorem uses the marked instruments, not an independent geometric-clock replacement. The audit gives the required affine second moment and strictly quadratic Bayes

risk. All hypotheses of the general theorem have now been verified directly from the primitive kernels. □

Theorem 7.2 (Noncommuting quantum instruments with report-selected stopping)**.** *Let $D_q \geq 2$, $\tau_* = I/D_q$, $0 < \lambda, \beta < 1$, and let A be a finite set of unitaries U_a. Let μ be normalized Lebesgue measure on the Hilbert–Schmidt unit ball of traceless Hermitian matrices. Put*

$$E_H = I + \beta H, \quad K_H = E_H^{1/2}, \qquad \mathcal{L}_a(H)\rho = \lambda U_a K_H \rho K_H U_a^* + (1-\lambda)\operatorname{tr}(E_H\rho)\tau_*.$$

This is a positive completely positive instrument, normalized after integration. Choose fixed Borel hazard functions $s_a(H) \in [0,1]$ with $\int s_a \, d\mu > 0$. Stop with probability $\alpha s_a(H)$ after the report. A bounded informationally complete audit gives Theorem 5.2 in dimension $d = D_q^2 - 1$. The effects and feedback unitaries need not commute. The observer is classical and has no uncharged quantum memory.

Proof. For a Hermitian matrix $\|H\|_{\mathrm{op}} \leq \|H\|_{\mathrm{HS}}$, so the Hilbert–Schmidt unit ball used here lies in the operator-norm unit ball. The effects satisfy $E_H \succ 0$, $\operatorname{tr} E_H = D_q$, and $\int E_H \, d\mu = I$. The report likelihood is $\ell(H,\rho) = \operatorname{tr}(E_H\rho) \in [1-\beta, 1+\beta]$. It genuinely depends on ρ. The raw map has the Kraus operation $\rho \mapsto K_H \rho K_H$, followed by a unital depolarizing channel and the selected unitary, a weak measurement with feedback rather than a parameterization by a preassigned Fisher matrix. Its normalized output has spectral floor $(1-\lambda)/D_q$. Let K be the density body with that floor and let D have half that floor. Then K has a positive affine buffer in D. The raw positive map sends the whole physical density base into K.

Take an informationally complete finite positive operator-valued audit and its probability coordinates. For example, a finite collection of sufficiently small positive perturbations of scalar identity effects spanning all Hermitian directions, together with a residual positive effect, suffices. The audit is chosen only after commitment and yields a single classical outcome. Its probability vector is an affine injective chart on trace-one matrices. In an orthonormal chart of its affine image the full squared categorical loss has the form (4), with all sampling variance retained. Linear instruments give the affine closure on every such coordinate. This does not assert a simultaneous measurement of noncommuting observables.

For acquired density it suffices to study normalized positive congruence before the final channel. For fixed $\rho \in D$ and a candidate positive density σ, the unique positive solution to $B\rho B = \sigma$ is

$$B_{\rho,\sigma} = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}.$$

Consequently the inverse effect on the trace-D_q hyperplane is

$$E = \frac{D_q B_{\rho,\sigma}^2}{\operatorname{tr}(B_{\rho,\sigma}^2)}. \tag{33}$$

Indeed $\sqrt{E}$ is a positive scalar multiple of B, which cancels on normalization. This proves injectivity, surjectivity on the attained image and a smooth inverse. Both the physical input and the effects are confined to compact positive interiors; the pre-channel output also has a uniform positive floor. Hence inverse derivatives are uniformly bounded, and the forward Jacobian is bounded away from zero. The final depolarizing-unitary map is invertible on the trace-one affine space, with linear scaling λ. Since the actual likelihood is bounded above, each acquired row has a uniformly bounded full-dimensional density. The statement remains valid in the audit chart by equivalence of finite-dimensional linear norms. It includes the boundary of the effect ball; that boundary has reference mass zero.

It remains to verify overlap for *all* feedback choices. Write $V_0(\rho) = \|\rho - \tau_*\|_{\mathrm{HS}}$. At $H = 0$,

$$V_0(F_a(0,\rho)) = \lambda V_0(\rho)$$

uniformly in a, because all unitaries fix τ_*. Uniform continuity gives, on a sufficiently small ball of reports, $V_0(F_a(H,\rho)) \le \lambda V_0(\rho) + c\|H\|$. Its actual conditional probability is at least $(1-\beta)$ times its fixed positive μ-mass. Choose a small neighborhood W of τ_* and then a sufficiently long fixed string of these quiet report balls to enter W from every $\rho \in D$, under every sequence of feedback unitaries. This event has a positive lower probability independent of the feedback history.

At input τ_* the acquired image is a full ball about τ_*, the same ball for every a. The smooth inverse (33) implies that, for all inputs in a sufficiently small W, a common smaller ball remains in all these images and the corresponding actual densities have a uniform positive lower bound there. One further acquired report thus gives (18) for some finite b, ϵ, ζ. These are real acquired-report events; they are not hypothetical physical resets. Their probabilities, Jacobian bounds and drift constants may deteriorate as $\lambda \uparrow 1$ or $\beta \downarrow 0$.

Report continuity follows from the trace pairing: $\mathrm{TV}(\ell(\cdot,\rho)\mu, \ell(\cdot,\sigma)\mu) \le (\beta/2)\|\rho - \sigma\|_1$. The common grids follow from the full density bound. Finally $D_a 1(\rho) \ge (1-\beta)\min_a \int s_a \, d\mu > 0$. This allows, for example, stopping only when a specified measured report direction lies in a hemisphere. All conditions have been verified for the same general theorem, independently of the Gaussian realization. □

8. Actual weighted affine strata

The full-dimensional hypothesis can be replaced by finitely many visibly marked affine components. The visibility of a raw mark is part of the acquisition interface, not an uncharged retained state.

Theorem 8.1 (Marked singular acquired geometry)**.** *Replace the row-density hypothesis in Definition 5.1 by the following condition. A finite raw mark j has actual probability $w_j > 0$, independent of the incoming state and action, with $\sum_j w_j = 1$. Conditional on this mark, the normalized predictive output lies in a compact convex affine body $K_j \subset K$ of dimension d_j, and its density is at most Λ_j in that affine space, uniformly over a, u. Components of dimension zero are allowed and are single points. For each positive-dimensional component assume*

$$w_j \underline{h}_j \le D_{a,j} 1(u) \le w_j \overline{h}_j, \qquad 0 < \underline{h}_j \le \overline{h}_j \le 1. \tag{34}$$

Keep the full-domain buffer and feedback-overlap certificate for the total normalized instrument, and the integrated total hazard bound. Then, for all sufficiently large fixed total label budgets,

$$\mathcal{R}_{\mathrm{cp}} - V_\alpha \asymp \Xi_w(M), \qquad \mathcal{R}_{\mathrm{hom}} - V_\alpha \asymp \Xi_w(M), \qquad \Xi_w(M) = \sum_{j:d_j>0} w_j M^{-2/d_j}. \tag{35}$$

Constants are uniform in α, M, u_0 and may depend on the fixed marked experiment. More precisely, every policy's actual terminal mark mass is

$$\widehat{w}_j^{\pi,\alpha} = \alpha \mathbb{E}_\pi \sum_{t<\tau} D_{A_t,j} 1(X_t), \qquad \frac{\underline{h}_j}{\eta_+} w_j \le \widehat{w}_j^{\pi,\alpha} \le \frac{\overline{h}_j}{\eta_-} w_j. \tag{36}$$

No component is normalized to unit mass when forming the total risk.

Proof. The Bellman argument is unchanged: it concerns the total acquired kernel and a concave value on D. Its restriction to any affine K_j is $O(\alpha)$-Lipschitz. Give each atom a label. For each positive-dimensional component choose a relative common grid with m_j cells and keep its labels distinct from those of other components. Apply the preparation-injected fixed-point construction on the product of these relative cell cones. A component's unnormalized continuing measure has density at most $w_j \Lambda_j \mathbb{E}\tau \le w_j \Lambda_j/(\alpha\eta_-)$. Its continuing Jensen loss is therefore at most $C_j w_j m_j^{-2/d_j}$ by Lemma 5.4 in that affine space. Atomic cells have no Jensen or terminal quantization loss. The terminal grid loss is at most the cell squared diameter times its actual terminal mass, hence at most $C_j w_j m_j^{-2/d_j}$ after (36). Use one preparation label, the finitely many atom labels, and allocate fixed positive proportions of the remaining labels to the positive-dimensional components. This

suffices for the upper in (35); it is not an optimal allocation statement when the weights themselves vary with the budget.

The identity (36) follows by summing the actual stopped kernels over the actual surviving histories. Its bounds use (24) and (34). The unnormalized terminal measure on component j has relative density at most $w_j\Lambda_j/\eta_-$. After division by its *actual* mass, its conditional density is bounded by $\Lambda_j\eta_+/(\eta_-\underline{h}_j)$. For any M global readouts, their orthogonal projections to the component's affine span can only decrease squared distance. The union-of-balls lower in that span yields $c_j\widehat{w}_j^{\pi,\alpha}M^{-2/d_j}$. Sum the costs against the genuine marked mixture and use the lower mass bound. This holds for every full-history acquisition policy, so it proves both checkpoint and observer converses. A checkpoint grid allocated among the finitely many components proves its upper. No imaginary full-dimensional mass has been assigned to the singular parts. □

Corollary 8.2 (Quantum observations with changing acquired affine rank)**.** *After the normalized quantum primitive in Theorem 7.2, draw a visible fixed mark j of probability w_j and apply a trace-preserving unital conditional expectation $\mathcal{E}_j$ onto a matrix subalgebra. Include the identity map with positive weight. Scalar expectation is an allowed atomic component; diagonal and full matrix expectations give distinct positive dimensions. With hazards of positive conditional report mass in every component, Theorem 8.1 applies.*

Proof. The raw marked map is $w_j\mathcal{E}_j\mathcal{L}_a(H)$, positive and affine. Trace preservation gives exactly the stated mark probabilities. In the real trace-one affine spaces, $\mathcal{E}_j$ is a linear projection onto its image. Fubini's theorem bounds the pushforward density by the original bound times a fixed maximal fiber volume of the compact density body; this is finite. The image of the floored density body is a compact convex affine body with relative interior. A scalar image is a point. Unital positivity preserves the spectral floor. The common interior buffer is still taken in the full physical affine space.

During the quiet-report argument for acquired overlap, additionally require every mark to be the identity mark. This has actual conditional probability $w_{\mathrm{id}}^b > 0$, independent of the controller, and retains the previous common overlap with its true reduced mass. The total instrument therefore satisfies the same type of feedback certificate. Hazards with a fixed positive report-region mass give (34) using $\ell \geq 1-\beta$. All closure, continuity and computational interface statements remain finite dimensional.

The rank changing here is the affine dimension of the acquired component. The strictly positive physical channel does *not* create zero eigenvalues of the state. These two notions of rank must not be conflated. Earlier pure-state rank-changing instruments are independent realizations under their own hypotheses. □

9. Acquisition, precision and causal transport

Corollary 9.1 (Internal halt status is paid)**.** *When a physical stop mark must instead be retained as internal halt status before the final audit action, the constructive observer with m grid cells uses at most $1+2m$ labels: one preparation label, m continuing labels and m halting labels. The halting labels have only the audit action and the fixed terminal decoder. Consequently the same $M^{-2/d}$ excess rate holds for this internally halted architecture for $M \geq 3$, with the same expected raw-acquisition order.*

Proof. Duplicate every grid label into a continuing and a halting copy. Route the acquired report to its common cell, then use the actual stop mark to select its copy. The positive instrument and stopped physical law are unchanged. The initial label is never revisited. Choose $m = \lfloor (M-1)/2 \rfloor$ for the upper; the uniform finite-codepoint converse applies to every M-label internally halted observer. No claim of minimality for the displayed duplication count is needed. □

Corollary 9.2 (A separately charged hard cap)**.** *In Theorem 5.2, impose an external hard cap $H \geq 1$: after the Hth raw report terminate even when its natural mark says continue. The auditor, not the observer, retains the cap counter. Let $V_{\alpha,H}$ be the full-history optimum for this capped task, and put $\rho_H = (1-\alpha\eta_-)^H$. Then*

$$|V_{\alpha,H} - V_\alpha| \leq L\rho_H,$$

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{cp},H} - V_{\alpha,H} \leq CM^{-2/d}, \tag{37}$$

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{hom},H} - V_{\alpha,H} \leq CM^{-2/d} + 2L\rho_H.$$

The homogeneous program may ignore the cap until termination. In particular $H \geq C_0\alpha^{-1}\log(M+1)$ gives matching capped curves, with C_0 depending on the fixed certificates and d. The physical maximum is $H+2$ calls including preparation and audit. This is not an internal-clock $M-H$ law.

Proof. Run the capped and uncapped policy with the same primitive and private coins. They have identical committed outcomes unless the natural run survives H calls. Extend a capped policy arbitrarily after H to obtain an uncapped one. Conversely, truncate any uncapped policy. The risk differences are at most $L\rho_H$, uniformly in the full-history policy, proving the value comparison. Use the uncapped finite observer with its fixed readouts also at a forced stop. Its risk changes by at most $L\rho_H$, giving the second upper.

Before H, the total naturally stopped acquired law has density at most Λ/η_- by (29). The forced terminal contribution at H has density at most Λ: it is a submixture of last-call acquired rows, multiplied by a continuing mark probability at most one. The sum is therefore bounded by $\Lambda(1+1/\eta_-)$. Uniform quantization gives the checkpoint lower, and hence the observer lower, relative to each policy's actual Bayes risk. A global grid at the final full-history predictive state gives the checkpoint upper. The exponential tail proves the displayed sufficient cap scale. No lower of order ρ_H has been claimed; the cap approximation error need not be necessary. □

Proposition 9.3 (Finite precision under the actual stopped law)**.** *Fix a constructed finite observer and its ideal task. Require the implementation to terminate almost surely or assign an explicit bounded failure outcome. Suppose a primitive/report/arithmetic implementation admits a conditional coupling, until first mismatch, whose probability of any discrepancy at one live call is at most θ. Include action, stop-mark and label discrepancies, and give a separate terminal-audit discrepancy bound θ_{end}. Then its complete stopped scored law differs in total variation by at most*

$$\frac{\theta}{\alpha\eta_-} + \theta_{\mathrm{end}}. \tag{38}$$

For a fixed ideal program whose terminal readouts are the true terminal centroids, replacing each readout by a legal point at distance at most δ_{mout} adds at most δ_{mout}^2 to the ideal risk. With both changes the risk increase is at most

$$L\left(\frac{\theta}{\alpha\eta_-} + \theta_{\mathrm{end}}\right) + \delta_{mout}^2.$$

The result requires a coupling certificate; it does not follow by comparing a discrete floating-point report and a continuous report in total variation.

Proof. The probability that first mismatch occurs at live call $t+1$ is at most $\theta\mathbb{P}(\tau > t)$ under the ideal law. Sum this unconditional survival bound and add the terminal discrepancy. On the complement, the whole stopped transcript, private coins, commitment and audit agree. The coupling characterization of total variation proves (38); it also proves almost-sure termination of the implementation on the coupled good event, with the failure event handled by the declared interface. A bounded scored loss changes by at most its bound times the mismatch probability. Conditional projection at the ideal terminal label eliminates the linear readout-error term and

leaves the squared displacement. Apply the coupling comparison to the same approximated readout on both laws to obtain the combined bound. □

The coupling hypothesis must include an implemented termination guarantee or a defined bounded failure score. It cannot be used to silently assign a finite loss to a machine that runs forever. A sufficient primitive guarantee is a uniform implemented integrated stopping floor; a separately charged cap also suffices. For piecewise-defined cells, a useful certificate splits a call error into primitive-coupling failure, arithmetic failure, and the *unconditional actual* row mass within a certified cell-boundary neighborhood. Terminal decoding precision is quadratic here; an error in the stopping or sensing kernel is generally an $O(\mathbb{E}\tau)$ perturbation, not automatically quadratic.

Proposition 9.4 (Typed causal resource composition)**.** *Let a parameter-independent causal simulator implement a target experiment from a source experiment. Specify a causal strategy lift, legal-action map, report and stopping-mark transformation, its internal state, acquisition clock and a common protected task readout. Assume that for every target observer in the declared class the complete stopped scored law, including its fresh private randomization, is within ε of the target law, uniformly in each inaccessible parameter under comparison. If the target observer has M labels and the simulator has R internal states, the composed observer uses at most MR dynamic state pairs and has risk at most the target risk plus $L\varepsilon$. Program, workspace, clock and readout costs are charged separately.*

For successive simulators, complete-law errors add and state counts multiply. Expected physical call budgets compose under explicit conditional expected-call bounds; hard caps compose only under explicit hard conditional bounds. A reverse minimax lower requires a reverse simulator or an independent converse.

Proof. A pair of observer and simulator states determines the next legal lifted action, the causal report transformation and the next state pair. No true parameter or future protected audit is an input. Independent coins may be generated afresh but are part of the joint coupling. The bounded-loss comparison follows directly from the complete-law total-variation assumption. Compose the causal kernels and apply the triangle inequality for the two errors, conditioning on the first simulator's complete history. Conditional summation of raw call bounds proves the expected resource assertion; deterministic summation proves its hard-cap counterpart. The direction of the simulation is preserved in the risk inequality, which explains the last statement. □

9.1. **Description length is not the number of labels.** The Borel existence theorem alone does not assert a computable exact fixed point. The following effective addition does not require finding that fixed point or introducing a physical cap.

Proposition 9.5 (Fixed-label effective approximation)**.** *Suppose the known raw instruments have a computable finite report algebra generating each reference sigma field, with certified integrals of every matrix coordinate over its cells, a computable integrable operator-norm domination, computable task constants, and a computable dense legal readout net. Assume a known uniform transience bound T_0 for every legal table. For every Borel M-label observer and every $\xi > 0$, there is a homogeneous M-label table with rational action/routing probabilities on a finite algebra partition and computably specified legal readouts whose risk is within ξ. Certified enumeration finds a table below any computable target strictly above the optimum. Its computation need not be efficient.*

Proof. Approximate each of the finitely many Borel routing rows in L^1 of the finite dominating measure having density proportional to $\sum_{a,b}\|I_a^b(y)\|$. Conditional averages on the increasing finite algebra partitions converge in that norm; rational probability vectors approximate the resulting simple rows without losing positivity or the sum-to-one constraint. The action probabilities are approximated in their finite simplices. Thus their finite ensemble maps (P', Q') approach (P, Q) in

operator norm. Both tables remain legal, so their resolvents have norm at most T_0, independent of the approximation. The resolvent identity and the unit preparation mass give

$$\|d' - d\| \le T_0\|Q' - Q\| + T_0^2\|Q\|\,\|P' - P\|.$$

A dense computable legal readout net then gives risk convergence on the compact response base. For an ideal centroid readout its squared displacement cost is the term in Proposition 9.3; the remaining difference is a bounded actual-law comparison.

Every rational finite table in this enumeration has a computable actual risk: integrate its finite positive matrix entries with certificates, approximate the Neumann sum with a remainder controlled by (3), and evaluate its finite readout cost. Dovetail these interval evaluations. Since some enumerated table has risk strictly below a strictly relaxed target, eventually an upper interval endpoint verifies it. No equality test at an irrational optimum or rate of convergence for the unknown ideal routing partition is used. The truncated Neumann sum is an offline calculation; the executed experiment is not capped or initialized in stationarity. Exact membership in report-algebra cells belongs to this ideal computational interface; numerical classification requires the separate boundary-mass certificate of Proposition 9.3. □

The stored description contains, separately, action rows, routing partitions, continuing constants and terminal decoders. For M labels, a chart dimension d and b bits per constant, the two lists of vector constants alone cost at most $2dMb$ bits. A dense finite-alphabet routing table can cost $O(M^2|A|Jb)$ bits for J report bins; this is not subsumed by $\lceil \log_2 M \rceil$ dynamic label bits. Temporary report precision, evaluation workspace, time per primitive, offline interval integration, and the external cap counter are additional resources. The main rate does not optimize this entire computational frontier.

Corollary 9.6 (An attained common-task calibration and deficiency law)**.** *After the stopped core has committed its label, freeze further core controls. Perform one paid report of an independent prepared bit $U \in \{-1, 1\}$, erased with probability $a \in [0, 1]$, and then a common protected audit of the core, the bit and an unobserved calibration sign. For the squared product task, with calibration magnitude δ, the source-to-exact-bit interface has causal deficiency exactly $\varepsilon = a/2$. For all sufficiently large M, the unrestricted checkpoint and a homogeneous counted observer satisfy, on this same experiment,*

$$\mathcal{R}(\alpha, M, \delta, \varepsilon) - V_\alpha(u_0) \asymp M^{-2/d} + \delta^2 + \varepsilon. \tag{39}$$

The upper is implemented with at most $4m + 2$ labels when the core uses m labels. The physical expected call count is $\mathbb{E}\tau + 3$ including preparation, the extra bit report and the final joint audit. This construction does not assert a lower for arbitrary allowed error tolerances.

Proof. For the simulation upper, transform an erasure into a fair guess of U. The output is wrong with probability $a/2$ and otherwise the complete target transcript can be coupled exactly. For the lower take the two possible preparations $U = 1$ and $U = -1$. On erasure the parameter-independent simulator has the same information under both; its worst conditional mismatch is at least $1/2$. The protected audit verifies the true bit and the virtual bit report must be committed before that audit. Thus the uniform deficiency is exactly $a/2$.

Under the equal preparation prior the optimal bit response is its observed sign, or zero on erasure, with squared loss exactly $a = 2\varepsilon$. The invisible calibration alternatives $\pm\delta$ have midpoint response zero and squared minimax loss δ^2. They are independent of the core and of every allowed report. Conditional projection on the product experiment therefore adds their lower losses to the uniform acquired-core quantization lower. Optimizing core acquisition cannot remove either term, since the extra bit is reported only after core control is frozen. For the upper, keep the core label through three terminal bit lanes $-1, 0, 1$ and one pre-bit lane, plus the constant interface states. This uses no more than $4m + 2$ labels. Decode the core by its programmed terminal centroid, the bit as above, and the calibration coordinate by zero. Taking m a fixed positive fraction of M proves the

order in (39). The label count is sufficient, not asserted minimal. Bounded task constants include the bit and calibration ranges; at a forced cap the already charged tail term must be added. □

10. A completely specified binary Gaussian observer

The raw Gaussian geometry constants below are inherited from the earlier geometric-stop benchmark [14]. The report-dependent stopping table and its separately disintegrated terminal law give exact integral formulas for one different finite-budget observer. It is not claimed to be the optimal three-label program.

Take a binary hidden state, scalar Bernoulli audit, preparation $u_0 = 1/2$, and

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \quad f_0(y) = \varphi(y+1), \quad f_1(y) = \varphi(y-1), \quad s(y) = \mathbf{1}_{\{y \geq 0\}}, \quad \alpha = 1/2,$$

where φ is the standard normal density. A call reports first and transitions second. The incoming state is its probability of hidden value one. Take $K = [1/4, 3/4]$, $D = [1/8, 7/8]$ and buffer $r = 1/8$. The scalar audit has $g(u) = u(1-u)$ and loss bound one. There is one sensing action; additional sensors may be included in Theorem 7.1, but are not part of this explicit table.

Proposition 10.1 (Constants and a three-label risk formula)**.** *Put $B_0 = \log 7 + 2$. The class certificates can be chosen as*

$$\eta_- = \Phi(-1), \quad \eta_+ = \Phi(1), \quad \Lambda = \frac{4e^{B_0+2}}{\sqrt{2\pi}}, \quad b = 1, \quad \epsilon = \frac{e^{-(\log 3 + B_0)^2/8}}{\sqrt{2\pi}},$$

with ζ uniform on $[3/8, 5/8]$ and Φ the standard normal distribution function. A homogeneous observer with preparation label 0 and running labels $1, 2$ is specified by the programmed routing points $q_0 = 1/2, q_1 = 3/8, q_2 = 5/8$. It routes to label one when

$$y < t_s = -\tfrac{1}{2} \log(q_s/(1-q_s))$$

and to label two otherwise, regardless of the stop mark. Its optimal fixed terminal readouts and exact risk are given by (40) *below. Thus the continuous-report program is fully specified by finitely many elementary and normal-integral constants.*

Proof. The actual log odds before transition are $Z = \log(u/(1-u)) + 2Y$. Their law is a mixture of $N(\log(u/(1-u)) \pm 2, 4)$, so its density is bounded above by

$$(2\sqrt{2\pi})^{-1} \exp(-(|Z| - B_0)_+^2/8).$$

The inverse logistic Jacobian is $2 + e^Z + e^{-Z} \leq 4e^{|Z|}$; the final affine transition contributes a factor two. Maximizing $t - (t - B_0)_+^2/8$ gives $B_0 + 2$, proving the displayed Λ. On final states $[3/8, 5/8]$, the pre-transition log odds lie in $[-\log 3, \log 3]$. Every Gaussian mixture component there has density at least $(2\sqrt{2\pi})^{-1} e^{-(\log 3 + B_0)^2/8}$. The inverse logistic Jacobian is at least four. Including the factor two and the interval length $1/4$ gives the stated ϵ. The hazard mass is a mixture of $\Phi(-1)$ and $\Phi(1)$. These bounds cover all incoming states in D, not just the symmetric preparation. They are explicit conservative constants, not fitted estimates.

The normalized predicted state is $F(y, q) = 1/4 + \frac{1}{2} \operatorname{logistic}(\log(q/(1-q)) + 2y)$. Its crossing of $1/2$ gives the programmed threshold. To give the exact actual risk, put $J_{s1} = (-\infty, t_s)$ and $J_{s2} = [t_s, \infty)$, and for $i = 0, 1$ define

$$a_{sji} = \int_{J_{sj}} f_i(y)\, dy, \qquad b_{sji} = \tfrac{1}{2} \int_{J_{sj} \cap [0,\infty)} f_i(y)\, dy.$$

All of these are differences of the explicit normal distribution function. Form a 6×6 continuing matrix C with two-hidden-state blocks indexed by $0, 1, 2$: its row block zero is zero, and for $j = 1, 2$

$$C_{j,s} = P \operatorname{diag}(a_{sj0} - b_{sj0}, a_{sj1} - b_{sj1}).$$

Form a 4×6 terminal matrix Q with blocks $Q_{j,s} = P\,\mathrm{diag}(b_{sj0}, b_{sj1})$. Let $a = ((1/2, 1/2), 0, 0)$ and set

(40)

$$z = (I - C)^{-1}a, \quad d = Qz, \quad w_j = d_{j0} + d_{j1}, \quad r_j = \frac{d_{j1}}{w_j}, \quad R = \sum_{j=1}^{2} w_j r_j (1 - r_j), \quad \mathbb{E}\tau = \sum_{s,i} z_{si}.$$

The positive hazard proves convergence of the inverse series. Successive integration of the raw two-state instrument proves the occupation and terminal equations, so these are exact formulas for the continuous model. Neither a numerical posterior nor a simulation is used in the proof.

For orientation, direct floating-point evaluation gives

$$\mathbb{E}\tau \simeq 4.411179, \qquad R \simeq 0.223714,$$

$$(w_1, w_2) \simeq (0.071311, 0.928689), \qquad (r_1, r_2) \simeq (0.433640, 0.667231).$$

The corresponding continuing means are approximately $(0.310468, 0.660317)$ at running labels one and two. They differ from both the programmed routing points and the terminal readouts. The formulas, not these rounded numbers, define the program and its guarantee. In particular this explicitly programmed table is not substituted for the fixed-point optimum in the matching theorem. □

The unrestricted checkpoint's optimum in this same stopped model is exactly (31). With $M = 3$ its excess over $V_{1/2}(1/2)$ lies between the explicit union-of-balls bound using Λ/η_- and the squared mesh of a three-cell interval grid. The three-label observer above provides a concrete, separately computable online upper. We do not claim that its numerical risk equals the checkpoint optimum or the optimal online risk.

11. Relation to prior work and scope

Sequential vector quantization studies dynamic encoders of Markov sources; finite-state POMDP control and finite approximations study implementable restrictions of partially observed control [1, 17, 15, 2]. Those subjects are direct antecedents, not unrelated background. A hard total-label budget differs from a window of raw reports and from the number of cells used to discretize an infinite-dimensional belief. Our lower applies to every permitted finite decoder and every full-history acquisition strategy of the declared stopped experiment. Our upper pays its preparation label and does not assume a free precise posterior register.

Value-directed belief compression explicitly preserves information relevant to future decision quality rather than an arbitrary norm on a full belief [7, 8, 10]. The executable task chart here serves a related purpose. The new statement is not that task-directed compression is novel. The distinction is the common positive-instrument feasibility criterion, the separate continuing and terminal first-run measures, and the matched hard-state risk law under report-selected stopping. A lossless linear test closure by itself neither supplies the finite routing table nor proves its resource converse.

The connection between Bayes-risk error and Bregman divergence is classical in prior quantization [16]. The Jensen terms in Theorem 4.1 are instances of that convex geometry. Their new role here is dynamic: preparation mixing and terminal selection must be disintegrated before a single observer can be calibrated. The two finite counterexamples show exactly what is lost if either term is silently identified with a different conditional law.

Absorbing-control occupation measures are particularly close to the present proof. Dufour and Prieto-Rumeau show why characteristic balance equations can admit measures which are not occupations, and study additional conditions giving occupation identification and compactness [3, 4]. We do not claim that positive resolvents or occupation balance are new. Uniform transience on a finite predictive cone provides our explicit identification step. The support-function criterion

additionally enforces common report routing of all incoming physical ensembles with a fixed finite label architecture. It eliminates those routing kernels but does not make the remaining optimization convex.

The recent finite-memory analysis of Kim [6] compares an exact belief and a fixed-policy finite-history belief approximation along the same closed-loop execution, under quantitative forgetting and Wasserstein control. Its memory object is a finite input–output window, and its value comparison replaces the belief argument while holding that execution fixed. In contrast our stopped benchmark optimizes all acquisition strategies in the declared library and our converse constrains a hard finite output alphabet. Neither conclusion subsumes the other: an infinite-precision window need not have finitely many labels, and the present hypotheses do not prove unknown-kernel learning or arbitrary belief-space forgetting.

Recent partition-observation work of Díaz-Infante Velasco, Garcia and Minjárez-Sosa [9] normalizes class-restricted conditional measures, includes atomic and continuous components, and derives a discounted-control formulation with finite approximations. A finite observation partition there need not produce a finite predictive or controller state. Our relative-stratum theorem instead accounts for actual marked masses inside a fixed hard-label architecture and proves both its causal upper and all-decoder lower. It does not claim that measure-level normalization or mixed observation classes are new.

The finite affine quotient, perspective identities, acquired-mass curvature lemma, Gaussian and quantum raw calculations, and the preparation-injected construction for an *independent* geometric stop have been developed in earlier companion work [11, 12, 13, 14]. The last of these uses a single geometric stopped ensemble because independent termination does not bias the recurrent row in the same way. We retain it separately and prove all ingredients needed here. The present extension permits report-dependent support of termination, optimizes the finite sensing/hazard library, identifies three distinct conditional means, adds the preparation defect for arbitrary label reuse, and proves global stopped feasibility before deriving its geometric consequences. The source lineage and editorial companions are not mathematical premises hidden from the main article.

11.1. **Scope of the completed statements.** The exact criterion applies to known finite affine predictive closures with the declared Borel interfaces and uniform transience. It is a necessary and sufficient recursive realizability test in that class; it is not an intrinsic scalar dimension formula for every raw experiment. The sharp result further assumes actual mass, global legal cells and acquired feedback overlap. These certificates are verified for nontrivial controlled classes, but are not proved necessary. Their constants need not remain bounded when the overlap or hazard floor degenerates.

Stopping is optimized through a fixed finite library with a common scale α and a uniform integrated floor. Unrestricted stopping under only an expected-time constraint can deliberately concentrate the acquired law, as Remark 4.4 shows. The theorem does not erase that distinction. Unknown-kernel learning, efficient global synthesis, nondominated infinite-dimensional physical states, arbitrary countable singular degenerations and the full optimal program/workspace/time frontier are not consequences proved here. None is inferred by replacing a finite stopped argument with infinite-horizon notation.

Appendix A. Two stability certificates are genuinely different

The matched stopping proof uses acquired feedback overlap and a Bellman oscillation bound. It does not identify these with either a uniform one-step curvature contraction or a same-report pathwise contraction. The following quantitative contrasts explain the older certificates' different roles.

Proposition A.1 (A large curvature envelope with pathwise contraction)**.** *In a binary positive hidden-state instrument let the transition be symmetric with crossover* $1/100$*, and let a binary report*

have probabilities $f_0(1) = 1/200$, $f_1(1) = 1/2$, *with complementary probabilities for report zero. On the operating interval* $[1/100, 99/100]$, *the uniform scalar one-step Bayesian perspective-curvature envelope exceeds* 6. *Nevertheless each report update followed by transition is a strict contraction in log-odds projective distance, with factor at most* $49/50$.

Proof. Let p be the incoming probability of state one, $\ell_y(p) = (1-p)f_0(y) + pf_1(y)$, and $F_y(p) = 1/100 + (49/50)pf_1(y)/\ell_y(p)$. Direct differentiation of the perspective potential gives

$$\tfrac{1}{2}\frac{d^2}{dp^2}\sum_y \ell_y(p)F_y(p)^2 = (49/50)^2 \sum_y \frac{(f_0(y)f_1(y))^2}{\ell_y(p)^3}.$$

At $p = 1/100$ its value is the rational number

$$\frac{387792606414808000000}{61181567631019341199} > 6.$$

Thus the product bound based on this uniform one-step envelope is not summable. A report multiplies odds by a positive scalar, so it is a log-odds isometry. For the symmetric transition with crossover a, differentiation of its log-odds map yields a Lipschitz bound $1 - 2a$; its maximum is at equal odds. At $a = 1/100$ this gives $49/50$. This does not exclude more refined multistep or localized curvature certificates. The exact statement is that this particular uniform one-step curvature route fails while the pathwise route succeeds. □

Proposition A.2 (Curvature contraction without uniform pathwise contraction)**.** *There is a positive finite-state report instrument whose integrated perspective-curvature coefficient is* $\omega < 1$, *but for which every finite block admits a positive-probability report word inducing the identity predictive update. Therefore no strictly contracting uniform pathwise block certificate holds on any nontrivial metric domain.*

Proof. With probability $\omega \in (0, 1)$ report an idle symbol and apply the identity map to the physical probability vector. Otherwise use positive report functionals m_y with $\sum_y m_y = e$, and apply the measure-and-prepare map $z \mapsto m_y(z)(1, v_y)$ with fixed interior v_y. Multiply these latter maps by $1 - \omega$. All likelihoods are positive on the simplex interior and the integrated map is normalized. The perspective potential is

$$\omega\|u\|^2 + (1-\omega)\sum_y m_y(1,u)\|v_y\|^2.$$

Its second term is affine, so its half-Hessian is exactly ωI. But an all-idle word of length k has probability $\omega^k > 0$ and leaves every predictive state unchanged. Such a word rules out a strict uniform same-report block contraction. The certificate comparison is exact. This finite-report example is not claimed to satisfy the full-density assumption of Theorem 5.2; the two propositions isolate stability implications, not additional rate realizations. □

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